MAST20026 Real Analysis Semester 2 2025

Assignment 3

Due: 11 September, 11:59pm

M= a mark associated with applying an appropriate method

J= a mark associated with using correct justification

A= a mark associated with a correct answer or intermediate step

N= a mark associated with good notation or writing

Please see the assignment instructions on the Assignment 3 page on the LMS, where you accessed this file. To ease submission, please prepare assignment solutions for each question on its own page. This document is 2 pages in length.

PART A (*2 marks* total)

Go to the article *On proof and progress in mathematics* linked from the Assignment 3 page on the LMS. Read it and write a short reflective response (around half a page) that considers the following questions.

- Was there anything about the reading that you found surprising?
- Focusing on sections 3 and 4, what (if anything) resonated with you? Does Thurston's discussion of communication and proof correspond with your journey with understanding mathematics?
- What do you think progress in mathematics looks like?

1 mark for demonstrating thoughtful engagement with the task.
1 mark for clearly communicated response (the response is well written, and conveys thoughts concisely).

PART B (*23 marks* total)

1. $\epsilon - M$ **definition**. *(8 marks)*

   (a) Let (f_n) be the sequence defined by

   $$f_n = \frac{n^2 - 4n}{n^2 - 2}$$

   for $n \in \mathbb{N}^+$. Use the $\varepsilon - M$ definition of convergence to show that $(f_n) \to 1$.

   (b) Let (g_n) be the sequence defined by

   $$g_n = \frac{e^{-\frac{1}{n}} + 1}{1 + n^2}$$

for $n \in \mathbb{N}^+$. Use the $\varepsilon - M$ definition of convergence to show that (g_n) does not converge to 1.
Remark: You can use properties of e^{-x} that you are familiar with. In particular, it is a monotone decreasing function with $e^0 = 1$.

Solution.

(a) Let $\epsilon > 0$.
Let $M = \max\left(\left\lceil \frac{12}{\epsilon} \right\rceil, 2\right)$.
Let $n \in \mathbb{N}^+$ such that $n > M$.
Note that if $n > M$, then in particular, we have

1M: Demonstrated understanding of the structure of an epsilon - M proof

- $n \geq 2$
- $n \geq \frac{12}{\epsilon}$

1A: Any valid M that makes the proof work

1J: A mostly correct sequence of inequalities / demonstration of the bounds on |f_n - L|

Then

$$\begin{aligned}
\left|\frac{n^2-4n}{n^2-2} - 1\right| &= \left|\frac{n^2-4n-n^2+2}{n^2-2}\right| \\
&= \left|\frac{-4n+2}{n^2-2}\right| \\
&\leq \frac{|-4n|+2}{|n^2-2|} && \text{(triangle inequality)} \\
&= \frac{4n+2}{n^2-2} && (n^2 > 2 \text{ since } n \geq 2) \\
&\leq \frac{4n+2n}{n^2-2} && (\text{ since } n \in \mathbb{N}^+) \\
&= \frac{6n}{n^2-2} \\
&\leq \frac{6n}{n^2-\frac{1}{2}n^2} && (\frac{1}{2}n^2 \geq 2 \text{ since } n \geq 2) \\
&= \frac{12}{n} \\
&\leq \frac{12}{12/\epsilon} = \epsilon && (n \geq \frac{12}{\epsilon})
\end{aligned}$$

1J: A completely correct sequence of inequalities / demonstration of the bounds on |f_n - L|, with key steps justified (in particular, any use of the triangle inequality, or an use of n>M)

We have shown: if $n > M$ then

$$\left|\frac{n^2-4n}{n^2-2} - 1\right| < \epsilon$$

Therefore $f_n \to 1$.

(b) Negating the definition of convergence, we note that we need to show:

$$\exists \epsilon > 0, \ \ \forall M \in \mathbb{N}^+, \ \ \exists n > M, \ \ |g_n - 1| \geq \epsilon. \quad (**)$$

Define $\epsilon = \frac{1}{2}$. (Other ϵ work, depending on the chain of inequalities produced)
Let $M \in \mathbb{N}$.

1M: A demonstrated understanding of the structure of a proof for "not converges to 1" looks like.
(namely, following the structure of a proof of (**))

1A: Any valid specification of n and epsilon that works for the proof presented.
(eg. epsilon = 1/2, n=M+1 is used in the solutions, but others could work too)

Let $n = M + 1 > M$. (Actually, any $n > M$ will work.)
Then $n > M \geq 1$ and

1J: A mostly correct sequence of inequalities / demonstration
of the bounds on |g_n - 1|

1J: A correct sequence of inequalities / demonstration of the bounds on |g_n - 1|, with key steps justified
(eg. use of bound on e^(-1/n), justify when absolute values get removed, and when n>M is used)

$$
\begin{aligned}
|g_n - 1| &= \left| \frac{e^{-1/n} + 1}{1 + n^2} - 1 \right| \\
&= \left| \frac{e^{-1/n} - n^2}{1 + n^2} \right| \\
&\geq \frac{|1 - n^2|}{1 + n^2} && (0 \leq e^{-1/n} \leq 1, \;\; n^2 > 1) \\
&= \frac{n^2 - 1}{1 + n^2} && (n > 1) \\
&\geq \frac{1 - 1/n^2}{1/n^2 + 1} \\
&\geq \frac{1}{1/n^2 + 1} && (0 < \frac{1}{n^2} < 1) \\
&\geq \frac{1}{1 + 1} && (\frac{1}{n^2} \leq 1) \\
&= \frac{1}{2} = \epsilon
\end{aligned}
$$

We have shown that $(**)$ is true, so $\lim_{n \to \infty} g_n \neq 1$.

1M: Demonstrated understanding of how to prove that A is not bounded (eg. by demonstrating understanding of the definition of bounded or unbounded).

1J: Demonstrated understanding of Archimedean Principle, and explicitly mentioned by name when used in the proof.

1J: Proof mostly correct, with no incorrect statements.

1N: The proof was well-written (in terms of layout), and the key ideas were easy to follow.

2. **More on boundedness.** (*4 marks*) Consider the following subset of $\mathbb{R}$.

$$A = \{x \in \mathbb{Q} \mid (\exists n \in \mathbb{N})\ x = n^3\}$$

Using the Archimedean Principle (any version) to prove that A is not bounded above in $\mathbb{R}$.

Solution.
Roughly: the main idea is that for all $n \in \mathbb{N}$, $n^3 \in A$, and $n \leq n^3$, so if there's a bound on A, then there's a bound on $\mathbb{N}$, contradicting the Archimedean property.

Here's a possible proof.
Proof:
Assume that A is bounded. (Contradiction Premise)
Therefore $\exists K \in \mathbb{R}\ \ \forall x \in A\ |x| \leq K$. (Definition of bounded)
Choose such a K.
Let $n \in \mathbb{N}$.
Note that $n^3 \in \mathbb{Q}$ so $n^3 \in A$.
Therefore $n^3 < K$, since A is bounded
Therefore $n < K$, since $n < n^3$.
So we have shown $\forall n \in \mathbb{N}\ \ n \leq K$. So $\mathbb{N}$ is bounded.
But by the Archimedean Property (Form I), $\mathbb{N}$ is not bounded.
$\mathbb{N}$ is bounded and $\mathbb{N}$ is not bounded is a contradiction.
We conclude that A is not bounded.

Alternative solution
We can also do a direct proof.
Proof:
We can show directly that A is not bounded above. This means we need to show

$$\forall K \in \mathbb{R},\ \ \exists a \in A,\ \ (a > K).$$

Let $K \in \mathbb{R}$
Case 1 ($K \leq 0$):
Let $K \leq 0$.
Define $a = 1$.
Clearly, $a = 1^3 \in \mathbb{Q}$, so $a \in A$.
Moreover $a = 1 > 0 = K$.
So there exists $a \in A$ such that $a > K$.

Case 2 ($K > 0$)**:**
Let $K > 0$.
By the Archimedean Property (version 2), there is a natural number $n \in \mathbb{N}$ such that $n - 1 < K \leq n$.
Define $a = (n+1)^3$.
Note that $a \in A$ since $n + 1 \in \mathbb{N}$.
Also $a = (n+1)^3 > n \geq K$.
So again, there exists $a \in A$ such that $a > K$.

Combining both cases we have shown: $\forall K \in \mathbb{R} \ \exists a \in A \ (a > K)$.
Therefore, A is not bounded above.

3. **Subsequences.** (*11 marks*) Let (f_n) be the sequence given by

$$f_n = \sin(n), \text{ for } n \in \mathbb{N}^+.$$

(a) Prove that (f_n) has a convergent subsequence.

(b) Can you explicitly construct a convergent subsequence? If yes, then define such a subsequence. If no, then explain why you were unable to do so – be precise about what was difficult.

(c) Show that there exists a subsequence (f_{n_k}) of (f_n) with the property that $f_{n_k} \geq \frac{1}{\sqrt{2}}$ for all $k \in \mathbb{N}^+$.

(d) Prove that (f_n) diverges.

Solution.

1A: State a correct bound on (f_n), and conclude (f_n) has a convergent subsequence.

(a) Using the properties of sin, we have

$$\forall n \in \mathbb{N}^+ \quad -1 \leq \sin(n) \leq 1$$

1J: Explicitly state when Bolzano - Weierstrass is used

So (f_n) is a bounded sequence.
By the Bolzano-Weierstrass theorem, (f_n) has a convergent subsequence.

1A: Yes or No, with a clear reason (either construct the subsequence, or state clearly why it is difficult to do so).

(b) The question asks if the student can construct an explicit convergent subsequence. While a student might potentially be able to do so with some difficult, I would expect most students to answer: "no, I was not able to construct a convergent subsequence".

The difficulty lies in trying to find precise integers n_k such that $sin(n_k)$ converges, since the period of sin is 2π is irrational.

Note in particular, that we can't just pick $n_k = 2\pi n$, for example, since $2\pi n$ is not a natural number.

(c) For each $k \in \mathbb{N}^+$, consider the interval

$$I_k = \left(\frac{\pi}{4} + 2k\pi, \frac{3\pi}{4} + 2k\pi\right) \subseteq \mathbb{R}^+$$

Note that for all $x \in I_k$, $\sin(x) \geq \frac{1}{\sqrt{2}}$. This is just the usual properties of the values of sin.
Observe that each interval I_k has length:

$$\frac{3\pi}{4} + 2k\pi - (\frac{\pi}{4} + 2k\pi) = \frac{\pi}{2}$$

Comment: I don't expect that most students will mention the Archimedean Principle here. It's enough for students to note the interval I_k has length greater than 1 so there must be a natural number in it.

1M: A valid approach to proving the statement

1J: Clearly defined subsequence / clearly defined how to construct the subsequence / demonstrated existence of such a subsequence

By the Archimedean Principle, there exists $n_k \in \mathbb{N}^+$ such thhat

$$n_k - 1 < \frac{\pi}{4} + 2k\pi < n_k$$

1J: Clear explanation for why sin(n_k) ≥ 1/sqrt(2)

Using the above inequality and $1 < \frac{\pi}{2} < 2$, we have

$$\frac{\pi}{4} + 2k\pi < n_k < \frac{\pi}{4} + 2k\pi + 1 < \frac{3\pi}{4} + 2k\pi$$

1N: Overall presentation of proof / construction was easy to follow, and key ideas could be understood.

so $n_k \in I_k$.

Therefore, for each $k \in \mathbb{N}^+$, we have found an element $n_k \in \mathbb{N}^+$ such that $\sin(n_k) \geq \frac{1}{\sqrt{2}}$. So $(f_{n_k}) = (\sin(n_k))$ is a subsequence of (f_n) with the property that $f_{n_k} \geq \frac{1}{\sqrt{2}}$ for all $k \in \mathbb{N}^+$

(d) Via a similar argument to part (c), we can construct a subsequence (f_{m_l}) such that for all $l \in \mathbb{N}^+$, $f_{m_l} \leq -\frac{1}{\sqrt{2}}$ by instead having m_l be the natural number in the intervals

1M: A valid approach to the problem — eg. noting that the argument in (c) can be repeated to construct another subsequence that must converge to something else (if it did converge)

$$\left(\frac{5\pi}{4} + 2l\pi, \frac{7\pi}{4} + 2l\pi\right) \subseteq \mathbb{R}^+$$

Comment: This is either from the divergence criteria theorem, or the small theorems before it (depending on the version you use)

Now suppose (for contradiction) that $f_n \to L$ for some $L \in \mathbb{R}$.
Then from a theorem in class, every subsequence of (f_n) must also converge to L.
From the subsequence (f_{n_k}) with the property that $f_{n_k} \geq \frac{1}{\sqrt{2}}$ for all $k \in \mathbb{N}^+$, and the result from class relating to inequalities of limits, we have

1J: Theorems from class relating to subsequences and convergence / divergence (eg. Divergence Criteria Theorem, theorem relating to inequalities)

$$\lim_{k\to\infty} f_{n_k} = L \text{ and } f_{n_k} \geq \frac{1}{\sqrt{2}} \implies L \geq \frac{1}{\sqrt{2}}.$$

But using the subsequence (f_{m_l}) with the property that $f_{m_l} \leq -\frac{1}{\sqrt{2}}$ for all $l \in \mathbb{N}^+$, we have

$$\lim_{l\to\infty} f_{m_l} = L \text{ and } f_{m_l} \leq -\frac{1}{\sqrt{2}} \implies L \leq -\frac{1}{\sqrt{2}}.$$

1J: A mostly correct argument with the key ideas included and no incorrect statements

But $L \geq \frac{1}{\sqrt{2}}$ and $L \leq -\frac{1}{\sqrt{2}}$ is a contradiction.
So we conclude that (f_n) does not converge to L for any $L \in \mathbb{R}$.
Therefore (f_n) diverges.

1N: Overall presentation / proof was easy to follow and key ideas could be understood.

Note: There are some other ways to do this proof. Some other options are:

- We could construct a subsequence that diverges (by alternating selecting terms from each subsequence). The apply the Divergence Criteria Theorem to conclude the original sequence diverges.
- We could even argue that the sequence is not Cauchy.